线性代数应该这样学

[Linear Algebra Done Right]

1. 向量空间的定义、基本性质
2. 线性相关、张成、基、维数的定义
   有限维向量空间的基本理论
3. 引入线性映射（主要结果：线性映射的零空间的维数加上值域的维数等于定义域的维数）
4. 多项式部分理论——理解线性算子所必需
5. 引入本征向量——将线性算子限制到更小到子空间上来研究
   复向量空间上本征值存在性到简洁证明
   复向量空间上的线性算子关于某个基有上三角矩阵
   实向量空间上的线性算子都具有 1 维或 2 维的不变子空间
   奇数维实向量空间上的线性算子都有本征值
6. 内积空间的定义、基本性质
   标准工具（规范正交基、格拉姆-施密特正交化过程以及伴随.....）
   利用正交投影解极小化问题
7. 谱定理（刻画了本征向量可以组成规范正交基的线性算子）
   正定算子、线性等距同构、极分解以及奇异值分解
8. 引入极小多项式、特征多项式以及广义本征向量
   （主要成果：用广义本征向量来描述复向量空间上的线性算子，可证几乎所有通常要使用 Jordan 形来证明的结果）
   （复向量空间上的可逆线性算子都有平方根，复向量空间上的线性算子都有 Jordan 形.....）
9. 实向量空间上的线性算子（核心）
   2 维不变子空间弥补了此类算子可能无本征值的不足，得到与复向量空间类似的结果
10. 利用特征多项式给出迹和行列式的定义（前面定义特征多项式时并未使用行列式）
   复向量空间上，定义的另一种陈述方式：迹是所有本征值之和；行列式是所有本征值之积（两种情况都计重数）
   利用极分解和自伴算子的刻画导出多重积分的换元公式

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- A. 复数
- B. 向量空间的定义
- C. 向量空间的定义
- D. 子空间
- E. 和与直和
- F. 习题

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- B. 复系数
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- A. 不变子空间
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- C. 上三角矩阵
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- E. 实向量空间的不变子空间
- F. 习题

第6章 内积空间
- A. 内积
- B. 范数
- C. 规范正交基
- D. 正交投影与极小化问题
- E. 线性泛函与伴随
- F. 习题

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- D. 正算子
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- G. 习题

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- B. 分块上三角矩阵
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- D. 矩阵的行列式
- E. 体积

符号索引

索引

1.0 向量空间

1.1 复数(complex number)

1) $i = \sqrt{-1}$
2) $i^2 = -1$
3) 复数集 C：$C = \{a + bi：a, b \in R\}$
4) 实数集 R
5) R(表示 C 或 R)中的元素称——标量(scalar)

1.2 向量空间的定义

- $F^n = \{(x_1, \dots, x_n): x_i \in F, i = 1, \dots, n\}$
- F^n 中的元素$x = (x_1, \dots, x_n)$看作一个始于原点的箭头——向量(vector)
- F^n 上的加法：通过相应坐标相加定义两个元素的和
- F^n 上的乘法：F 中元素与 F^n 中元素的乘法——标量乘法
- 向量空间(vector space)：带有加法和标量乘法的集合 V
- V 上的加法(addition)：指一个函数，把每一对每一对$u, v \in V$，都对应到 V 的一个元素$u + v$
- V 上的标量乘法(scalar multiplication)：指一个函数，把任意$a \in F, v \in V$，对应到 V 的一个元素$uv \in V$

Linear Algbra Done Right

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Chapter 1 Vector Space

Vector spaces are a generalization of the description of a plane using two coordinates, as published by *Descarts* in 1637.

1. Vector Space

Linear algebra is the study of linear maps on finite-dimensional vector spaces.

Learning properties of the complex numbers

- basic properties of the complex numbers
- R^n and C^n
- vector spaces
- subspaces
- sums and direct sums of subspaces

1.1 R^n and C^n

1.1.1 Complex Numbers

R: the set of real numbers

i: denote a square root of -1, that obeys the usual rules of arithmetic

1.1.2 Lists

1.1.3 F^n

1.1.4 Digression on Fields

1.1.n Exercises

1.2 Definition of Vector Space

1.2.n Exercises

1.3 Subspaces

1.3.1 Sums of Subspaces

1.3.2 Direct Sums

1.3.n Exercises

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1. The geometry of linear equations

n linear equations & n unknowns → row picture & *column picture → matrix form, AX = b

$$\begin{cases} a_{11}x + a_{12}y = b_1 \\ a_{21}x + a_{22}y = b_2 \end{cases} \leftrightarrow \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \leftrightarrow x\begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} +\cdot\, y\begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$\begin{cases} a_{11}x + a_{12}y + a_{13}z = b_1 \\ a_{21}x + a_{22}y + a_{23}z = b_2 \\ a_{31}x + a_{32}y + a_{33}z = b_3 \end{cases} \leftrightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \leftrightarrow x\begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \end{bmatrix} + y\begin{bmatrix} a_{12} \\ a_{22} \\ a_{32} \end{bmatrix} + z\begin{bmatrix} a_{13} \\ a_{23} \\ a_{33} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

2. Elimination with matrices

elimination → success/failure → back-substitution, elimination matrices, matrix multiplication → A to U(upper triangular)

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & b_{22} & b_{23} \\ 0 & b_{32} & b_{33} \end{bmatrix} \rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & b_{22} & b_{23} \\ 0 & 0 & c_{33} \end{bmatrix} \quad 3 \text{ pivots}(\mathrm{a}_{11}, \mathrm{b}_{22}, \mathrm{c}_{33} \neq 0)$$

3. Multiplication and inverse matrices

4. Factorization into A = LU

5. Transposes, permutations, spaces R^n

6. Column space and nullspace

7. Solving Ax = 0: pivot variables, special solutions

8. Solving Ax = b: row reduced form R

9. Independence, basis, and dimension

10. The four fundamental subspaces

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